

Federated Learning for Cloud Cybersecurity: Privacy-Preserving Collaborative Threat Intelligence

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Abstract—Artificial intelligence is becoming more prominent to detect and mitigate cyber threats in a cloud computing environment. Nevertheless, conventional centralized machine learning architectures demand massive amounts of sensitive information, posing a level of high privacy risk and governmental issues. Federated learning serves as one of the promising paradigms capable of training the model collaboratively and maintaining the decentralization of data at the same time. This study aims to suggest a privacy-sensitive cybersecurity system that combines federated learning, zero-trust access control, and secure DevSecOps lifecycle in cloud environments. The suggested architecture will combine the distributed client training, encrypted model aggregation, and the continuous threat detection pipelines to enhance security without affecting user privacy. To illustrate the capability of federated updates to improve the detection of anomalies in distributed systems, a machine learning threat detection workflow is developed. Another concept presented in the study is a layered zero-trust cloud security model that will be used to enforce identity verification, network segmentation and application level protection. The findings suggest that federated solutions minimize risks of exposing data and do not compromise high detection rates and scaling. The article adds a unified architectural framework that enables privacy-saving AI to comply with contemporary cloud cybersecurity needs.

Index Terms—Federated Learning, Cloud Security, Privacy-preserving AI, Zero Trust Architecture, DevSecOps, Cyber Threat Detection.

I. INTRODUCTION

Cloud platforms have been critical to contemporary digital infrastructures and they support scalable storage, distributed applications and real-time analytics. Concurrently, cloud services have led to heightened complexity of the cybersecurity threats such as distributed denial of service attacks, insider threats, and advanced persistent attacks. Artificial intelligence and machine learning have become common logical to automatize threat detection and response processes, yet the conventional centralized training process presupposes access to sensitive datasets that might be against privacy laws or organizational standards [1], [2].

Federated learning (FL) is a solution to this dilemma wherein decentralized hardware/organizations can jointly learn

a common model without the need to send raw data to a centralized repository. Every participant conducts local training and uploads model change only, which enhances privacy and compliance [3], [4]. Recent literature points out the possibilities of FL in cloud and edge security, which proves to be more beneficial in collaboration between distributed networks, and maintains confidentiality [1], [5].

In spite of these benefits, current studies tend to address individual aspects like threat sniffers or privacy-enhancing systems, which do not provide a comprehensive architectural design. Not many studies integrate federated learning with zero-trust security and secure DevSecOps frameworks that are necessary to implement AI models in a safe environment on real-world clouds [6], [7].

The paper suggests a unified model that combines federated learning with the process of zero-trust access, machine learning threat detection pipelines, and secure-by-design DevSecOps. The contributions of this work are as follows:

- Federated learning cloud system to distributed cybersecurity intelligence.
- A workflow of zero-trust access with the validation of identity and devices at all times.
- A secure aggregation machine learning threat detection pipeline.
- A safe DevSecOps lifecycle of privacy-saving AI deployment.

The rest of this paper will be organized in the following way. Section II is a review of related work. The third section (III) presents the proposed architecture. Sections IV-VII contain the workflow of zero-trust, threat detection pipeline, DevSecOps lifecycle and the model of layered security.

II. RELATED WORK AND LITERATURE REVIEW

The recent literature highlights privacy-preserving artificial intelligence as a prerequisite in the cloud cybersecurity. Albshaier et al. [1] present a systematic review of the challenges in federated learning, putting a spotlight on the issues of scalability and the security of the models. As noted by Alejandra and

Elena [6], privacy-preserving AI models can improve security analytics in distributed cloud environments.

Camila et al. [2] show that federated learning allows sharing threat intelligence collaboratively without revealing sensitive information of the organisation. In the same way, Elganzori et al. [8] introduce AI classification models to achieve privacy-aware cybersecurity, where they demonstrate higher accuracy in detecting anomalies. Goswami [7] discusses the use of privacy-sensitive AI in the governance of cloud security.

Federated learning solutions are also relevant to enterprise decision system because of lower risk of data leakage that can occur during training [9]. Hussain [3] suggests privacy-saving designs based on decentralized learning plans, whereas Iyengar et al. [10] emphasize the use in digital forensics and cyber investigations.

The latest research broadens the application of federated learning into geopolitical areas in which cross-organizational interaction is needed without endangering sovereignty [5]. Multi-cloud applications also promote scalability due to model training being distributed among a variety of infrastructures [11]. Studies on IoT-based smart cities have shown that federated models enhance dynamism in cyber threat detection without violating privacy [12], [13].

According to Rodriguez [14] and Timofte et al. [15], incorporating federated learning with advanced cybersecurity analytics is a way to enhance resilience to changing attacks. Table I compares the literature of privacy-preserving federated learning approaches.

TABLE I
LITERATURE COMPARISON OF PRIVACY-PRESERVING
FEDERATED LEARNING APPROACHES

Ref	Focus Area	Key Contribution	Limitation
[1]	FL Security Review	Challenges & AI Opportunities	Limited architecture design
[2]	Cybersecurity FL	Collaborative threat intelligence	No zero-trust integration
[12]	IoT Threat Detection	Real-time FL detection	Infrastructure complexity
[15]	Cybersecurity FL	Privacy-preserving analytics	Deployment challenges

The literature has shown that there is a significant research gap, as most studies discuss either privacy or cybersecurity individually and do not give an integrated secure cloud architecture.

III. PROPOSED FEDERATED LEARNING CLOUD ARCHITECTURE

The architecture proposed will present a security-sensitive architecture in which distributed client nodes will be used to jointly train cybersecurity models without jeopardizing security measures.

The given federated learning cloud design is aimed at supporting collaborative cybersecurity intelligence on the base of ensuring the strict privacy of data and compliance with regulations. The classical centralized computational models of machine learning are based on the centralization of sensitive

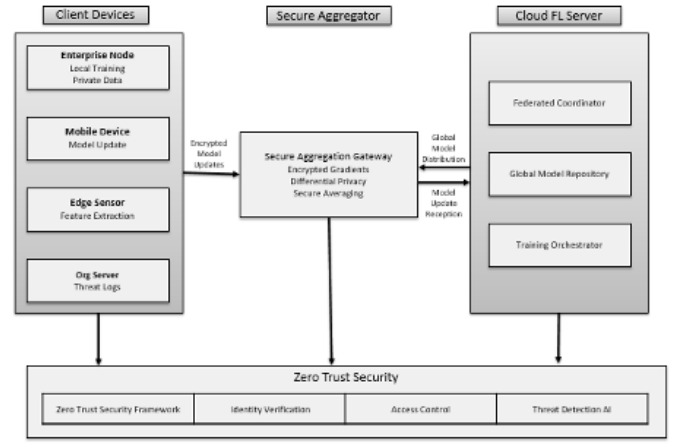


Fig. 1. Federated learning cloud architecture for privacy-preserving cybersecurity applications.

information in one repository, which poses the risk of data leakage, insider-attack, and the breach of compliance. Conversely, the architecture suggested is decentralized, with training being conducted in various client nodes, which enables local data to be stored within the limits of an organization but make contributions to a global model.

The system has three key layers, as shown in Fig. 1, which include the client device layer, the secure aggregation gateway, and the cloud federated learning server. The enterprise nodes, mobile endpoints, edge sensors, and organizational servers are all client devices. Individual model training is done by each participant with its own local cybersecurity information (e.g. network traffic logs, behavioral analytics, or threat intelligence feeds). Rather than sending raw data, only encrypted model parameters are sent to the aggregation server. The method is consistent with the previous findings that show that decentralized learning is less threatening to privacy, and it does not impair analytical performance [1], [3].

The secure aggregation gateway will be necessary to prevent interruption and manipulation of model update. Differential privacy, encrypted gradients, and secure averaging are some of the techniques that will safeguard the fact that the individual client contribution cannot be reconstituted by malicious actors. The federated averaging process can be mathematically defined as shown in (1):

$$w_{t+1} = \sum_{k=1}^K \frac{n_k}{n} w_{t+1}^k \quad (1)$$

where w_{t+1} represents the aggregated global model weights, K is the number of clients, n_k is the data size of client k , and w_{t+1}^k represents the local model weights uploaded by client k . These tools increase the levels of confidentiality and do not impair the effectiveness of learning and collaboration [12], [15].

A federated coordinator coordinates rounds of model training on the cloud, and storing the aggregated parameters on the global model repository. The training organizer issues updated

models to clients, and learning cycles are conducted continuously. Zero-trust security controls are implemented on each component and, therefore, there is identity verification and policy controls prior to any communication. Such a stacked implementation guarantees the resistance to an unauthorized access attempt and the ability to scale its deployment across multi-cloud infrastructure [11].

IV. ZERO TRUST ACCESS WORKFLOW



Fig. 2. Zero trust access workflow.

Zero-trust architecture assumes that implicit trust is not permitted to any user, device or system component of the network, no matter where it is located. The suggested workflow will incorporate identity verification, device posture verification, and policy-directed authorization to gain access to resources of federated learning. Fig. 2 is a workflow (step by step) that starts with a user or client request.

Upon a request being initiated, multi-factor authentication and credential validation is carried out by the identity verification module. The token access is only issued once verification is successful, and unauthorised users are not able to communicate with the system. The next step is device trust evaluation, which evaluates functioning system integrity, patch, and endpoint setup. The steps are key since compromised devices are able to inject malicious updates to federated models and this may cause attacks of model poisoning [6].

The zero-trust policy engine analyzes the contextual aspects based on behavioral abnormalities, physical location, and past activity examples. Risk scoring algorithms allow, restrict or deny access. This continuous appraisal framework is specifically relevant in decentralized AI settings whereby threats change very quickly. Studies point toward the resistance of AI pipelines to cybersecurity, identifying that the incorporation of zero-trust principles into the system can greatly increase the level of cybersecurity [7], [13].

When approved, the users are allowed to access federated learning resources including model APIs or safe aggregation services. Monitoring processes are on-going, and the behavior of the session is evaluated by using continuous check systems, which send warning messages in case of any abnormal activity. Feedback loops enable the monitoring systems to revise the policy rules dynamically, so as to enforce adaptive security. The architecture minimizes the attack surfaces and facilitates collaboration across organizations preserving privacy by integrating zero-trust controls throughout the working process.

V. MACHINE LEARNING THREAT DETECTION PIPELINE

The machine learning threat detection pipeline is a system that incorporates real-time security analytics with federated training.

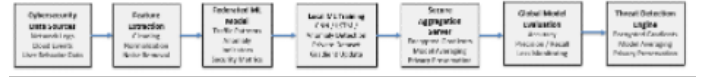


Fig. 3. Machine learning threat detection pipeline.

Machine learning threat detection pipeline converts unprocessed cybersecurity data into intelligence that can be acted on by a set of organized processing steps. Given that the data sources used in the analyzed case, including network logs, cloud activity logs, and user behavior indicators, are preprocessed to eliminate noise, normalize features, and maintain the same formatting, as depicted in Fig. 3. The indicators extracted by feature engineering include traffic abnormalities, aberrant login behaviors and security measures.

Federated client training at local nodes Sub-networks are trained at federated client nodes, where models like convolutional neural network (CNN) and long short-term memory (LSTM) models are trained on patterns in private datasets. These models produce gradient updates which represent learnt threat signatures without giving away sensitive information. Secure aggregation involves combining updates of various clients with the aim of developing a global model that will be able to identify threats in a wide range of environments.

Accuracy, precision, recall, and loss monitoring are evaluation metrics that give the information about model performance. Table II provides sample values of performance of federated intrusion detection models. According to previous research, federated solutions are as accurate as centralized systems, but much more privacy-safe [12], [14].

The last part of the piping is threat detection engine that produces intrusion warnings, risk rating, and automated response measures. The continuous learning loops help the models to be flexible in responding to new threats as they learn new updates provided by the client nodes. This scalability ability is especially useful in cloud scenarios where attack patterns are changing at a very fast rate. The pipeline will offer privacy-conscious and scalable cybersecurity monitoring through the combination of federated learning and cutting-edge analytics.

TABLE II
MODEL PERFORMANCE METRICS

Model	Accuracy	Precision	Recall
CNN-Based IDS	94%	92%	90%
LSTM Anomaly Model	96%	94%	93%

Federated aggregation generates a world model that can identify advanced threats to distributed infrastructures. Studies show that the federated learning model can attain similar performance to those of centralized models without losing data privacy [12], [14].

VI. SECURE-BY-DESIGN DEVSECOPS LIFECYCLE

The secure-by-design DevSecOps lifecycle is a concept that incorporates the security practices in all phases of the software and AI development process. DevSecOps integrates controls



Fig. 4. Secure-by-design DevSecOps lifecycle.

on security during the design and implementation phase, unlike the case of traditional development pipelines where security is implemented in the final phases of testing. Fig. 4 shows a lifecycle of infinity-loop relating operations and development activities.

Threat modeling and setting security requirements of federated learning systems is a part of the planning phase. To reduce vulnerabilities at production stage, static analysis tools and dependency checks are used during the coding to detect vulnerabilities. The build stage encapsulates federated learning artifacts into the secure containers such that they are compatible with cloud environments.

Testing involves vulnerability testing, penetration testing and testing machine learning models against adversarial attacks. These measures are obligatory since hacked models may bring biases or security issues. The release phase entails the checks and approvals of compliance to ascertain adherence to the privacy rules [13].

The deployment has applied zero-trust literatures, which mandate secure setup and regulated entry to cloud resources. Throughout the operations, real-time monitoring of the system behavior is possible with the help of run-time protection and threat logging. Ongoing surveillance promotes ongoing feedback in the planning phase, which forms an adaptive security cycle. This lifecycle also guarantees that the models of federated learning are secure during their operational life and promote the rapid innovation.

VII. ZERO TRUST CLOUD SECURITY MODEL

Zero-trust cloud security model proposes a layered model of securing federated learning environments. The various layers deal with particular security issues but are interoperable with other elements. The uppermost layer deals with user and machine interaction whereby only those parties that are



Fig. 5. Zero trust cloud security model.

authenticated are allowed to begin communication as seen in Fig. 5.

The identity and access management layer implements the policies of multi-factor authentication and roles-based access control. Continuous verification is used to ensure that privileges are automatically refined depending on the risk assessment. Under this, the network security layer has micro-segmentation and encrypted communication channels to ensure, in the future, the attackers will not move laterally.

The application layer contains federated learning services, threat detectors and DevSecOps pipelines. The architecture can isolate the application components hence minimizing its effects due to possible breach. The data protection layer is a layer that uses the methods of differential privacy, encryption, and anonymization to protect sensitive information in the course of training and storage. Lastly, the cloud infrastructure layer offers scalable resources including containers and virtual machines on which the system is based.

This is a multi-layered model, which is consistent with the recent studies on defense-in-depth approaches to cloud cybersecurity [8], [7], [15]. The architecture helps secure the collaboration in distributed environments by integrating identity management, network protection and privacy-preserving AI.

VIII. DISCUSSION

The suggested framework has a number of benefits over the conventional centralized cybersecurity solutions. To begin with, federated learning deals a huge blow to the possibility of data leakage by storing sensitive data in local environments. This method contributes to adhering to the regulations and enhances the cooperation between the organizations that cannot exchange raw data because of privacy issues.

Second, the adoption of zero-trust workflows improves the state of access control by verifying the users and devices continuously. Ongoing surveillance systems are used to detect suspicious activities and cause automatic reaction, which lowers the chances of insider threats. It has been shown that the integration of federated learning and zero-trust systems en-

hances the resiliency of systems to sophisticated cyberattacks [6], [5].

Third, DevSecOps lifecycle makes certain that the element of security is integrated throughout the development and deployment process. Automated testing and monitoring improve operational risks and allow the threat detection model to be updated quickly. Nevertheless, there are still some challenges such as the overhead in communication when aggregating models and the fact that adversarial attacks can be made against federated updates.

The innovations in the future can include sophisticated versions of differential privacy, secure multi-party computation, and blockchain-based trust to make the process more transparent. In spite of these issues, the framework offers a scalable basis of privacy-conserving cybersecurity on the current cloud environments.

IX. CONCLUSION

This study introduced a comprehensive framework that had federated learning, zero-trust security principles, and secure DevSecOps practices to overcome privacy issues in cloud cybersecurity. The suggested architecture allows training machine learning models distributedly and securing sensitive data with the help of encryption and policy enforcing measures.

Threat detection pipeline based on machine learning shows how federated learning can be used to safely and effectively increase the detection of anomalies without data centralization, and the layered zero-trust model adds more security to the system against unauthorized access. The DevSecOps life cycle also makes sure that security is a part of the development and deployment phases.

The next steps in work will be to enhance scalability by employing adaptive aggregation strategies and using real-time edge intelligence to improve the response to threats. With the further advancement of cloud environments, privacy-protective AI systems, including the one suggested in this paper, will become instrumental in facilitating safe and cooperative cybersecurity infrastructures.

X. FUTURE WORK

Future studies may extend the proposed federated learning framework for cloud cybersecurity by integrating distributed sensing infrastructures and scalable network resource management mechanisms. Techniques used in wireless communication systems may support efficient distributed data transmission and collaborative learning among edge and cloud nodes [16]. Furthermore, energy-efficient IoT sensing platforms could enable large-scale data collection from distributed environments while maintaining sustainable operation of edge devices [17]. Future research may also investigate privacy-preserving optimization algorithms, adaptive federated learning strategies, and secure communication protocols to enhance collaborative cybersecurity intelligence across heterogeneous cloud infrastructures.

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